

Investigation of Aeroelastic Stability on AGARD 445.6 Wing at Transonic Regime

Mustafa Ozcatalbas

Mechanical Engineering Department
TOBB Economy and Technology University
Ankara, Turkey
e-mail: mustafa.ozcatalbas@roketan.com.tr

Bulent Acar

Finite Element Analyses and Test Department
Roketsan Inc.
Ankara, Turkey
e-mail: bacar@roketan.com.tr

Sıtkı Uslu

Mechanical Engineering Department
TOBB Economy and Technology University
Ankara, Turkey
e-mail: suslu@etu.edu.tr

Abstract—Fluid-structure interaction problems (FSI) are too complex to solve analytically. Therefore, wind tunnel tests or computational methods are used to evaluate the integrity of aerospace structures. Performing wind tunnel tests for many design parameters in early stage of the design is very expensive. So, computational solid mechanics (CSM) and computational fluid dynamics (CFD) solvers which works coupled can be used to determine aeroelastic characteristics of the structure to reduce number of tests. In this study, flutter analyses of AGARD 445.6 wing are carried out using FSI method with commercially available CSM and CFD solvers. The procedure developed with the solvers is validated with the experimental data conducted by AGARD 445.6 standard aeroelastic configuration for dynamic response at subsonic and transonic free stream Mach numbers.

Keywords—fluid-structure interaction; AGARD 445.6; flutter; dynamic aeroelasticity

I. INTRODUCTION

Aeroelasticity is composed of aerodynamic, inertial and elastic forces which are best described by Collar triangle [1]. The combination of three mentioned forces can cause flutter phenomena that can be defined as unstable self-excited oscillation in which the structure gain energy from the air stream. After flutter is occurred, exponentially increasing oscillations lead to catastrophic structural failure. Since the flutter is an irreversible aeroelastic problem, it should be considered in the early stage of the aerospace structure design.

In transonic regime, flow field is nonlinear because of the moving shock waves caused by unsteady motion of the aeroelastic structure. The shock wave occurred in transonic flow field significantly reduces flutter speed which is called as “transonic dip” [2]. In this case, flutter velocities are overpredicted when linear potential theories are used to calculate unsteady aerodynamic loads [3-4]. Thus, nonlinear flow solvers based on Euler or Navier-Stokes equations which are coupled with structural deformity are used to have accurate flutter calculations [5].

Both analyses and tests should be accomplished to clear an aerospace structure from flutter in transonic regime. In preliminary stage of the design, bulk of aeroelastic analyses demand increases to minimize number of wind tunnel tests. Fluid structure interaction (FSI) analysis is one of the computational methods to determine flutter characteristics of a structure. Basically, FSI is a procedure which combines computational solid mechanics (CSM) and computational fluid dynamics (CFD) solvers and it enables them to exchange data with each other [6-7]. In FSI procedure for flutter calculations, aerodynamic forces, which are pressure and shear stress, are accepted as loading conditions for CSM solver, and aerodynamic grid of CFD is changed according to the displacement output of the deformed structure solved by CSM depended on corresponded aerodynamic forces. This computation loop continues until a criteria is satisfied or a steady-state condition is reached.

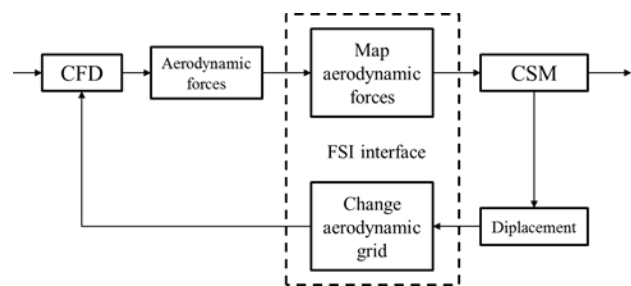


Figure 1. Two-way FSI flow diagram.

FSI can be provided by two different calculation algorithms: 1) strong coupling, 2) weak or loosely coupling. In weak coupling, data is exchanged at the end of every time step. On the other hand, strong coupling uses subiterations between time steps to synchronize CFD and CSM. Therefore, convergence increases as computation cost rises in strong coupling. Strong coupling is adequate for highly nonlinear aeroelastic systems, although loosely coupled FSI method is computationally efficient for linear flutter analyses [8-10].

This study focuses on vibrational behavior of AGARD 445.6 wing in transonic regime. There are many studies related to aeroelasticity calculations of AGARD 445.6 wing [11-15]. More recently, Goud et al. evaluated the effects of pressure, turbulent kinetic energy, and temperature on AGARD 445.6 wing at Mach 0.9 [16]. The objective of the study is to develop a loosely coupled FSI method with commercially available CFD-CSM solvers to investigate flutter onset of AGARD 445.6 wing. Also, the paper includes the validation of the results with available test data given by Yates [17].

II. MODEL DESCRIPTION

AGARD 445.6 weakened model #3 which was tested by Yates is used during the flutter analyses [17]. The geometrical properties of the wing are shown in Fig 2. The wing has the root and tip chord of 0.558m and 0.386m, and semi-span of 0.762m. It has taper ratio of 0.66, an aspect ratio of 1.65, and quarter-chord sweep angle of 45° . The wing has NACA 65A004 profile with no twist or curvature along its length.

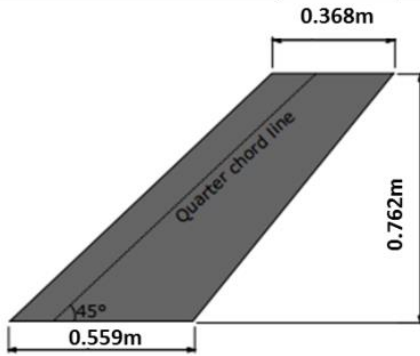


Figure 2. AGARD 445.6 wing planform.

A weakened AGARD 445.6 is made of laminated mahogany which has orthotropic material properties given by Yates [17].

TABLE I. AGARD 445.6 MATERIAL PROPERTIES

Material Property	Value
E1	3245.5 MPa
E2	416.2 MPa
G12	439.2 MPa
G13	439.2 MPa
G23	439.2 MPa
ν	0.31
ρ	381.9 kg/m ³

III. METHODOLOGY

Since FSI is a coupled process of different solvers; separate computational models are prepared for CSM and CFD in Abaqus [18] and Star-CCM+ [19] accordingly. The validity of both models is important to estimate flutter onset correctly. Therefore, CSM and CFD models are validated before the FSI analyses begin.

After the validation of the CSM and CFD models, steady-state CFD calculation of interested flow condition, which is used as initial condition for FSI analysis, is done. Then, vibration characteristics of the wing are evaluated and computation loop described in Fig. 1 is continued until undamped oscillation is observed. The above-mentioned approach of the study is summarized in Fig. 3 and further explained in following topics.

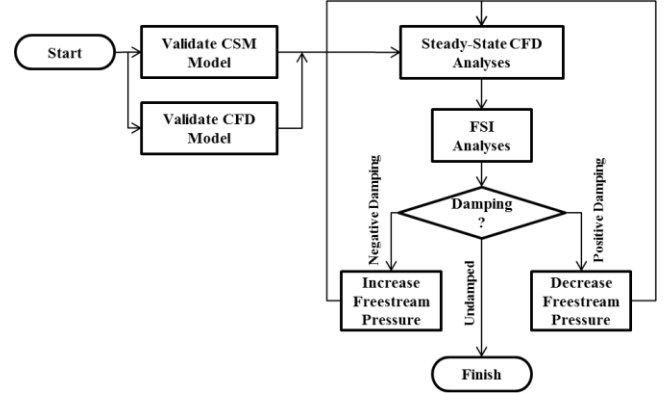


Figure 3. Flowchart for the approach of the study.

A. CSM Modeling and Simulation

Validation of CSM model is important to provide dynamic characteristics of the wing while dynamic implicit analyses are carried out by Abaqus during the FSI analyses. Finite element (FE) analyses done with Abaqus are performed to obtain natural frequencies, and mode shapes of the wing. These natural frequencies and mode shapes are used to validate CSM model.

The CSM model used in natural frequency and FSI analyses consists of 27270 linear hexahedral solid elements. There are 4 elements along the thickness. The FE model is given in Fig. 4.

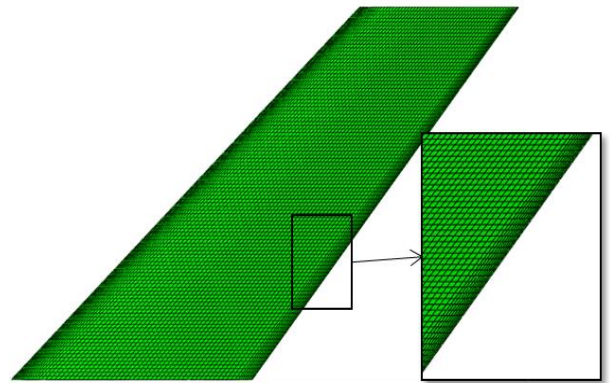


Figure 4. AGARD 445.6 FE model.

The calculated natural frequencies and mode shapes of weakened AGARD wing are in agreement with experimental results [17] and those computed by Kolonay [20], Lee and Batina [11].

TABLE II COMPARISON OF CALCULATED NATURAL FREQUENCIES (Hz)

	Mode-1	Mode-2	Mode-3	Mode-4	Mode-5
Present Study	9.30	39.36	48.93	95.41	123.18
Exp. [17]	9.60	38.17	48.35	91.54	118.11
Kolonay [20]	9.63	37.12	50.50	89.94	-
Lee & Batina [11]	9.6	38.17	48.35	91.54	-

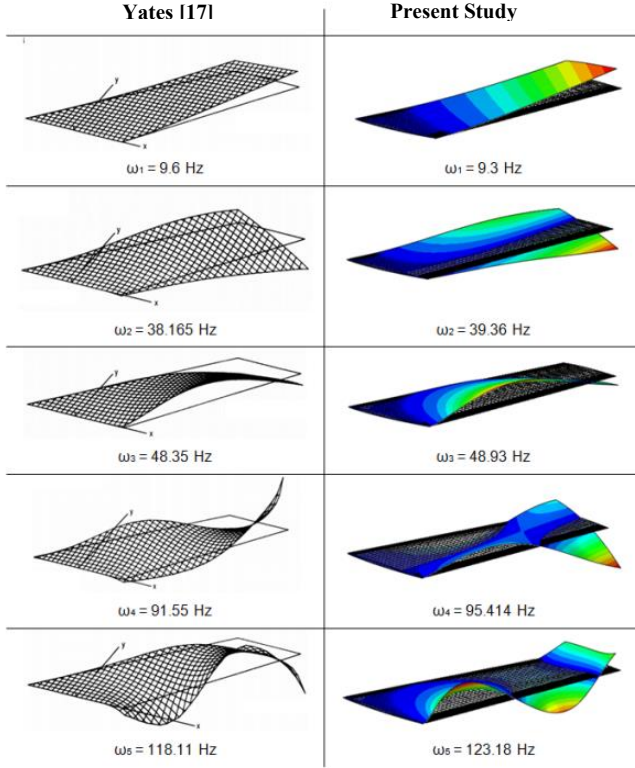


Figure 5. Comparison of calculated mode shapes.

B. CFD Modeling and Simulation

The fluid flow is assumed fully compressible and turbulent during the analyses. Therefore, Unsteady Reynolds Averaged Navier Stokes (URANS) approach with Spalart-Allmaras turbulent model has been adopted.

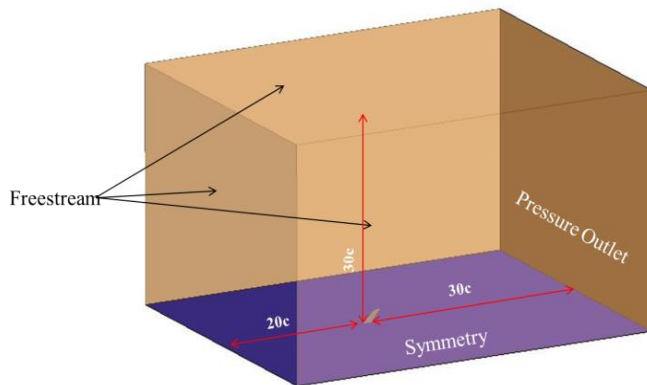


Figure 6. CFD domain.

The CFD simulations are carried out using commercially available Star-CCM+ which is based on finite volume method. Second order advection scheme is chosen for coupled solver. Besides, implicit second order forward time-stepping algorithm is used to discretize the unsteady terms in governing equations.

The computational domain is depicted in Fig. 6. No-slip boundary condition (BC) is applied to AGARD wing and flow conditions are defined in freestream BC. The optimum number of cells is determined as 1612446 after grid sensitivity analyses. Moreover, polyhedral volume mesh is preferred to increase convergence.

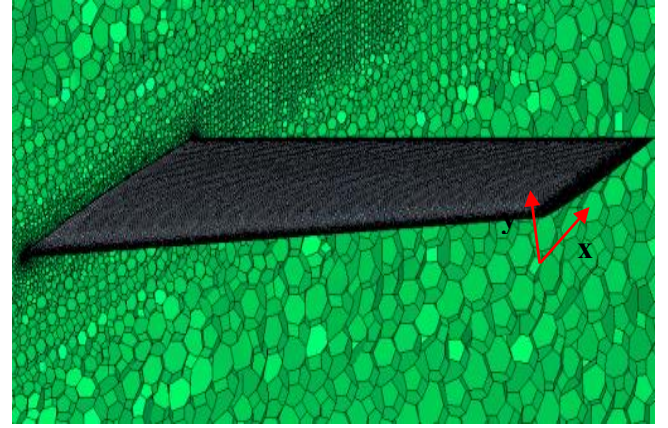


Figure 7. Computational domain of AGARD wing.

The steady-state CFD analyses of different flow conditions are compared with the studies conducted by Cai [21], Lee and Batina [11]. As shown in Fig. 8 and Fig. 9, the results agree well with the flow conditions of $M=0.85$, 5° angle of attack (AoA) and $M=1.141$, $AoA=0^\circ$.

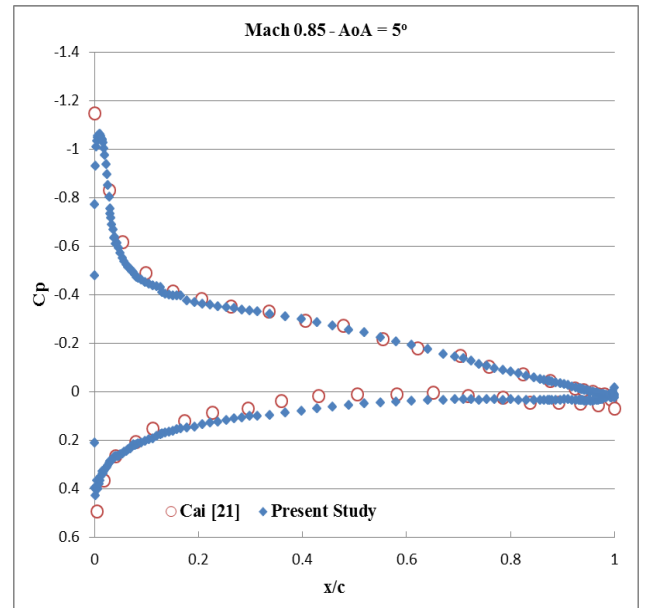


Figure 8. C_p comparison at 34% semi-span ($M=0.85$, $AoA=5^\circ$).

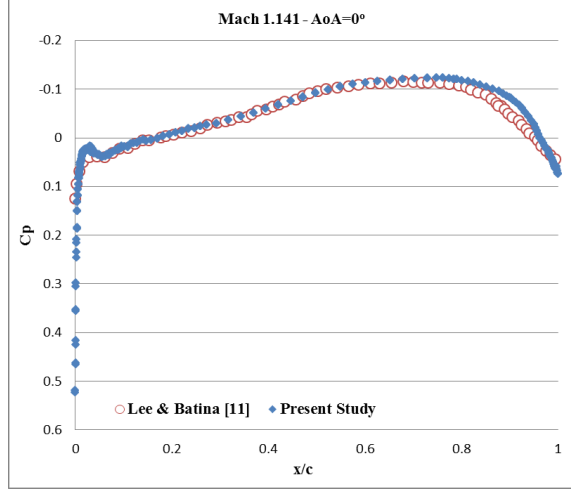


Figure 9. C_p comparison at 26% semi-span ($M=1.141$, $AoA=0^\circ$).

C. Flutter Analyses

Loosely coupled FSI method is chosen to perform flutter analyses. While the aeroelastic system is linear, this method is sufficient to provide interaction between CSM and CFD. Also, computation cost is reduced, in contrast to close coupling algorithm [10].

In the first step of the flutter analyses, first bending mode of the AGARD wing is excited. Therefore, time-dependent g-force along the y-direction as seen in Fig. 10 is applied to the wing during 0,025 s. After 0.025 s, the g-force is released and the wing vibrates freely in freestream.

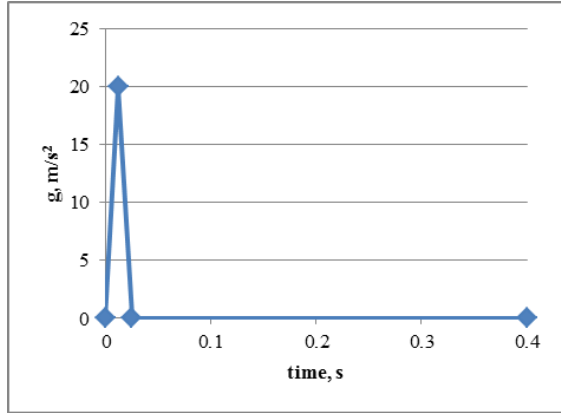


Figure 10. Time-dependent g-force.

During the FSI analyses to determine flutter onset of AGARD wing, constant coupling time interval of 4×10^{-4} s is selected. This time-step corresponds to sampling the period of the first torsional mode of dry wing in 60 points. Also, the transient analyses are carried for 0.4s.

The wind tunnel flutter test data given by Yates is presented in Table III. Flutter speed index (U_f) is a nondimensional number which defines the flow conditions where the undamped oscillations occur, and is given as:

$$U_f = \frac{U_\infty}{b\omega_\alpha\sqrt{v}} \quad (1)$$

where U_∞ is freestream velocity, b is half chord length, ω_α is first torsional mode frequency, and v is mass ratio represented as (2).

$$v = \frac{\rho_{wing} V_{wing}}{\rho_{air} V_{cone}} \quad (2)$$

where $V_{wing} = 0.0043584 \text{ m}^3$ and $V_{cone} = 0.13054 \text{ m}^3$ which is a truncated cone enclosing the wing [8].

TABLE III. EXPERIMENTAL FLUTTER DATA [13]

Mach	ρ (kg/m ³)	V (m/s)	U_f	ω/ω_α
0.499	0.4277	172.5	0.4459	0.5353
0.678	0.20818	231.4	0.4174	0.4722
0.96	0.06338	309	0.3076	0.3648
1.072	0.05512	344.7	0.3201	0.3617
1.141	0.07883	364.3	0.4031	0.4593

As to obtain flutter speeds, tip displacements of the wing are plotted as given in Fig.11. Damping characteristics of the aeroelastic system is evaluated, according to the magnitude change of the tip oscillations in time. And, the freestream velocities which make neither decaying nor growing oscillations are accepted as flutter speeds.

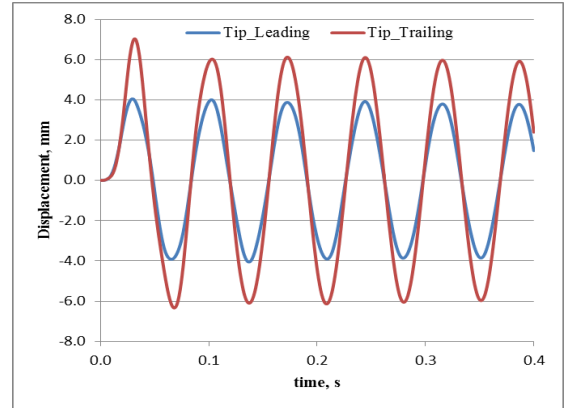


Figure 11. Vertical displacement for wing-tip leading and trailing edge displacements for undamped case $M=1.141$.

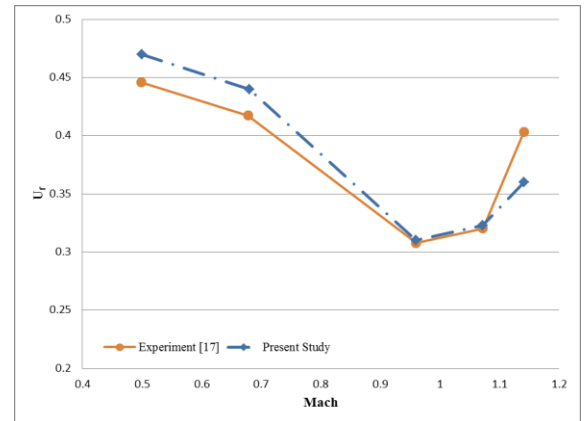


Figure 12. Comparison of calculated U_f with the experimental data.

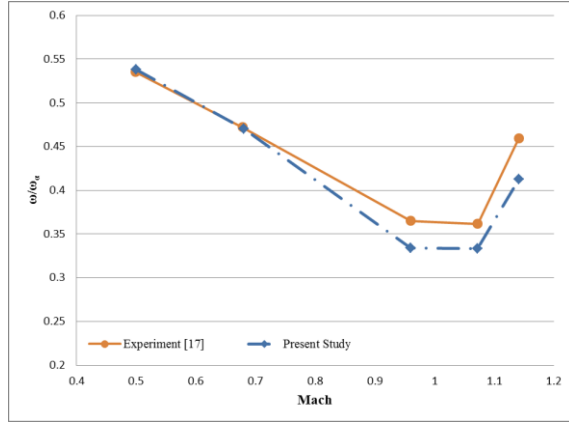


Figure 13. Comparison of calculated frequency ratio with the experimental data.

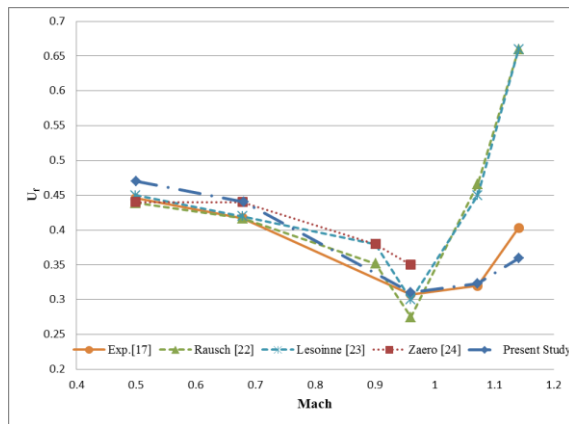


Figure 14. Comparison of calculated U_f with the experimental data and numerical studies.

IV. CONCLUSIONS

In this study, a loosely coupled FSI method which is achieved with commercial Star-CCM+ CFD and Abaqus FE solvers is presented to investigate flutter onset of AGARD 445.6 wing. The proposed simulations prove to predict dynamic aeroelastic characteristics of AGARD wing accurately in transonic flow conditions.

The calculated flutter speeds and frequency ratios are in good agreement with the experiment [17] and numerical studies [22-24]. The transonic dip is determined accurately. Furthermore, the presented U_f values are much closer to the experimental results than the previously conducted numerical studies, which are depicted in Fig. 15, especially when the Mach number is greater than 0.96. The research presented here considers the viscous effect while the other numerical studies use Euler equations. Hence, flutter speeds in high transonic regime are calculated more precisely, due to the increasing shear force effect in high speed flows [11].

As a next step, structural nonlinearities will be taken into consideration for an aeroelastic system. FSI method described in this study will be used to investigate the effect of nonlinearities on flutter.

REFERENCES

- [1] A. R. Collar, "The Expanding Domain of Aeroelasticity," *J. R. Aeronaut. Soc.*, pp. 613–636, 1946.
- [2] K. Isogai, "On the transonic-dip mechanism of flutter of a sweepback wing", *AIAA J.*, 1979.
- [3] D. M. Schuster, D. D. Liu, and L. J. Huttsett, "Computational aeroelasticity: success, progress, challenge", *J. Aircraft*, vol. 40, no. 5, pp. 843-856, 2003.
- [4] R. Yurkovich, "status of unsteady aerodynamic prediction for flutter of high-performance aircraft", *J. Aircraft*, vol. 40, no. 5, pp. 832-842, 2003.
- [5] L. Cavagna, G. Quaranta, and P. Mantegazza, "Application of Navier-Stokes simulations for aeroelastic stability assessment in transonic regime", *Computers & Structures*, pp. 818-832, 2007.
- [6] R. Kamakoti and W. Shyy, "Fluid-structure interaction for aeroelastic applications," *Prog. Aerosp. Sci.*, vol. 40, no. 8, pp. 535–558, 2004.
- [7] C. Sangeetha, "Fluid Structure Interaction on AGARD 445.6 wing at Transonic Speeds," vol. 2, no. 4, pp. 28–34, 2015.
- [8] E. Başkut and A. Akgül, "Development of a Closely Coupled Procedure for Dynamic Aeroelastic Analyses," *Sci. Tech. Rev.*, vol. 62, no. 2, pp. 30–39, 2012.
- [9] M. Akgul, "Static Aeroelastic Analysis of a Generic Slender Missile," Middle East Technical University, 2012.
- [10] STAR-CCM User Guide. CD-Adapco, 2015.
- [11] E. M. Lee-Rausch and J. T. Batina, "Calculation of AGARD Wing 445.6 Flutter Using Navier-Stokes Aerodynamics," 1993.
- [12] F. Thirifay and P. Geuzaine, "Numerical Simulations of Fluid-Structure Interaction Problems using MpCCI Franc."
- [13] G. S. Goura, "Time Marching Analysis of Flutter Using Computational Fluid Dynamics," 2001.
- [14] F. Liu, J. Cai, and Y. Zhu, "Calculation of Wing Flutter by a Coupled Fluid-Structure Method," *J. Aircr.*, vol. 38, no. 2, 2001.
- [15] X. Chen, "Numerical simulation of 3-D wing flutter with fully coupled fluid – structural interaction," vol. 36, pp. 856–867, 2007.
- [16] T. S. K. Goud, S. K. A, and S. S. Prasad, "Analysis of Fluid-Structure Interaction on an Aircraft Wing," vol. 3, no. 9, pp. 146–152, 2014.
- [17] E. C. Yates, "AGARD Standart Aeroelastic Configuration For Dynamic Response," Hampton, 1987.
- [18] ABAQUS, version. 2016, Dassault Systemes.
- [19] Star-CCM+, version. 12.04, Simulia Inc.
- [20] R. M. Kolonay, "Unsteady Aeroelastic Optimization in the Transonic Regime," Purdue University, 1996.
- [21] J. Cai, F. Liu, H. M. Tsai, and J. Reno, "Static Aero-elastic Computation with a Coupled CFD and CSD Method," in 39th AIAA Aerospace Sciences Meeting & Exhibit, 2001.
- [22] R. D. Raussch, J. T. Batina, H. T. Yang, "Three-Dimensional Time-Marching Aeroelastic Analyses Using an Unstructured-Grid Euler Method", *AIAA Journal*, vol. 31, no. 9, pp. 1626-1633, 1993.
- [23] M. Lesoinne and C. Farhat, "High order subiteration-free staggered algorithm for nonlinear transient aeroelastic problems", *AIAA Journal*, vol. 36, no. 9, pp. 1754-1757, 1998.
- [24] ZAERO Application Manual, ZONA Technology, Inc.